

# Richard Álvarez

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## Research Areas

Human–computer interaction; human-centered AI; computational media; AI-assisted creative tools.

## Education

B.A. in Film Aug 2020–May 2024  
Kenyon College Gambier, Ohio  
*Minor in History; Integrated Program in Humane Studies.*

## Publications and Posters

- [1] Álvarez, Richard Anthony. *A Retrieval-Augmented Film Recommendation System*. Poster, *IPHS 484: Senior Seminar*, Paper 34, Digital Kenyon, 2024. Digital Kenyon.
- [2] Álvarez, Richard Anthony. *Captivating Video Editing Secrets: Unsupervised Deep Learning and PySceneDetect Analysis of Creative Strategies*. *IPHS 484: Senior Seminar*, Paper 22, Digital Kenyon, 2023. Digital Kenyon.

## Professional Experience

**AI Solutions Engineer** Sep 2025–Present  
*Layer 7 Systems, Independent Contractor* Remote

Built and operated a stateful TypeScript, Bun, and Express MCP service connecting Retell voice agents with Autotask and Microsoft Graph/Teams.

Implemented identity and company resolution, contact-safe ticket access, authorization gates, ticket actions, technician routing, post-call processing, and human handoff.

Developed an authenticated React operations dashboard, maintained a 756-test regression suite, and deployed Docker services through GitHub Actions.

Analyzed 60 workflow recordings from 14 users and identified 176 moments suitable for AI assistance.

Built a Next.js and Neon/PostgreSQL learning platform for 34 trainees across four courses and 16 modules.

**Freelance Software and Web Developer** Oct 2024–Present  
*Richard Álvarez, Sole Proprietorship* Chicago, Illinois

Design and deploy responsive applications using Next.js, React, Node.js, Tailwind CSS, Stripe, PostgreSQL, authentication, and object storage.

Scope, document, deploy, and maintain client applications, hosting, and content workflows.

**IT Assistant** Sep 2023–Feb 2024  
*Kenyon College Library and Information Services* Gambier, Ohio

Supported workstation imaging, hardware deployment, classroom technology, and recurring maintenance scheduling.

## Selected Software and AI Projects

|                                        |           |
|----------------------------------------|-----------|
| Vigil                                  | 2025–2026 |
| Half-Time Draw Predictor               | 2026      |
| Chroma                                 | 2026      |
| GREasyVocab                            | 2024      |
| Bitset Game of Life                    | 2024–2026 |
| Kubrick-Style Script Generation        | 2024      |
| Rotten Tomatoes Sentiment Analysis     | 2024      |
| DV Tools                               | 2025      |
| Business and AI Training Platform      | 2026      |
| Workflow Capture and Analysis Pipeline | 2026      |
| Configurable Voice-Reception Template  | 2025      |
| Message-Box Auto-Attendant             | 2025      |
| VibeVoice Evaluation Prototype         | 2025      |

## Selected Professional Projects

|                                          |              |
|------------------------------------------|--------------|
| Joaquin Morales Cinematography Portfolio | 2024–2026    |
| Machine Television                       | 2024–2025    |
| BDPA HSCC Website Prototype              | 2025         |
| Personal Portfolio                       | 2024–Present |

## Professional and Community Service

### High School Computer Competition Judge

2024–2025

*BDPA*

*Chicago, Illinois*

Reviewed and scored student programming submissions for the 2024 and 2025 national competitions and completed formal judge training.

## Professional Certification

|                                |      |
|--------------------------------|------|
| CompTIA IT Fundamentals (ITF+) | 2024 |
|--------------------------------|------|

## Technical Capabilities

*Agent systems and orchestration:* OpenClaw custom agents, skills, memory, and tool integrations; MCP; tool calling; structured outputs; retrieval-augmented generation; context design; multi-agent delegation; human-in-the-loop controls.

*AI development environments:* Claude Code, Codex, GitHub Copilot and VS Code agent workflows; OpenAI, Anthropic, Gemini, and Retell platforms.

*Workflow automation and integration:* n8n, REST APIs, webhooks, Autotask, Microsoft Graph, Microsoft Teams, identity and permissions.

*Reliability and operations:* Docker, GitHub Actions, Linux/VPS operations, CI/CD, automated testing, validation, structured logging, observability, incident response, cost monitoring.

*Product and data engineering:* TypeScript, Python, Bun, Node.js, Hono, FastAPI, React, Next.js, Electron, PostgreSQL, Neon, SQLite FTS5, Zod.

*Systems and computational media:* C++20, CMake, FFmpeg, PySceneDetect, NLP and sentiment analysis, interactive media; Rust under active study.